

AS stoichiometry homework 4 · solutions and gas volumes

CIE 9701 AS Chemistry · 16 MCQ · 16 marks

Name: _____ Class: _____ Date: _____

Topics: 13 solutions and concentration · 14 gas volumes at room conditions. Later questions also use earlier skills (mole, empirical formula, hydrates, limiting reagent).

Circle **one** letter (A–D) for each question. Show working. Calculators and a Periodic Table may be used. For gas volumes, use $24 \text{ dm}^3 \text{ mol}^{-1}$ at room conditions (not $pV = nRT$), unless a question gives other conditions.

13 · Solutions and concentration

1. Sample X is added to water and made up to a total volume of 200 cm^3 . This gives a solution of 0.100 mol dm^{-3} HCl.

What is X?

- A. 10 cm³ of 1.00 mol dm⁻³ HCl
B. 30 cm³ of 0.90 mol dm⁻³ HCl
C. 50 cm³ of 0.40 mol dm⁻³ HCl
D. 100 cm³ of 0.30 mol dm⁻³ HCl

$$[1]$$

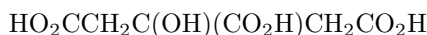
2. Which mixture will react to form exactly one mole of water?

	volume 2.00 mol dm ⁻³ H ₂ SO ₄ / cm ³	volume 1.00 mol dm ⁻³ NaOH / cm ³
A	250	500
B	250	1000
C	500	500
D	500	1000

$$[1]$$

3. Citric acid is found in lemon juice.

citric acid



Which volume of 0.40 mol dm^{-3} sodium hydroxide solution is required to neutralise a solution containing 0.0050 mol of citric acid?

- A** 12.5 cm^3 **B** 25.0 cm^3
C 37.5 cm^3 **D** 50.0 cm^3

$$[1]$$

4. Hydrogen peroxide, H_2O_2 , decomposes into water and oxygen when a suitable catalyst is added. 20.0 cm^3 of aqueous hydrogen peroxide decomposes to produce 600 cm^3 of oxygen at room conditions. What is the concentration of the aqueous hydrogen peroxide?

- A.** 5.00 mol dm^{-3}
B. 2.50 mol dm^{-3}
C. 1.25 mol dm^{-3}
D. $0.625 \text{ mol dm}^{-3}$

$$[1]$$

5. A washing powder contains sodium hydrogencarbonate, NaHCO_3 , as one of the ingredients.

In a titration, a solution containing 1.00 g of this washing powder requires 7.15 cm³ of 0.100 mol dm⁻³ sulfuric acid for complete reaction. The sodium hydrogencarbonate is the only ingredient that reacts with the acid.

What is the percentage by mass of sodium hydrogencarbonate in the washing powder?

- A** 3.0% **B** 6.0%
- C** 12.0% **D** 24.0%

$$[1]$$

6. The juice of one lemon reacts completely with 120 cm^3 of 0.50 mol dm^{-3} sodium carbonate solution.

The formula of citric acid is $\text{HOOCCH}_2\text{C}(\text{OH})(\text{COOH})\text{CH}_2\text{COOH}$.

No sodium carbonate is left unreacted.

What is the amount of citric acid in one lemon assuming that it is the only acid in the sample?

- | | | | |
|----------|----------|----------|----------|
| A | 0.02 mol | B | 0.04 mol |
| C | 0.06 mol | D | 0.09 mol |

$$[1]$$

14 · Gas volumes at room conditions

7. A sample of propane, C_3H_8 , with a mass of 9.61 g is completely combusted in an excess of oxygen under room conditions.

Which volume of carbon dioxide gas is produced?

- A 4.89 dm^3 B 5.24 dm^3
C 14.7 dm^3 D 15.7 dm^3

[1]

8. A mixture of 10 cm^3 of methane and 10 cm^3 of ethane was sparked with an excess of oxygen. After cooling, the residual gas was passed through aqueous potassium hydroxide.

All gas volumes were measured at the same temperature and pressure.

Which volume of gas was absorbed by the alkali?

- A 15 cm^3 B 20 cm^3
C 30 cm^3 D 40 cm^3

[1]

9. In an experiment, 0.100 mol of propan-1-ol is burnt completely in 12.0 dm^3 of oxygen, measured at room conditions. What is the final volume of gas, measured at room conditions?

- A 7.20 dm^3 B 8.40 dm^3
C 16.80 dm^3 D 18.00 dm^3

[1]

10. What is the total volume of gas produced, measured at room conditions, when 0.010 mol of anhydrous magnesium nitrate is completely decomposed by heating?

- A 240 cm^3 B 480 cm^3
C 600 cm^3 D 720 cm^3

[1]

11. The carbonate of an s-block element is reacted with an excess of hydrochloric acid.

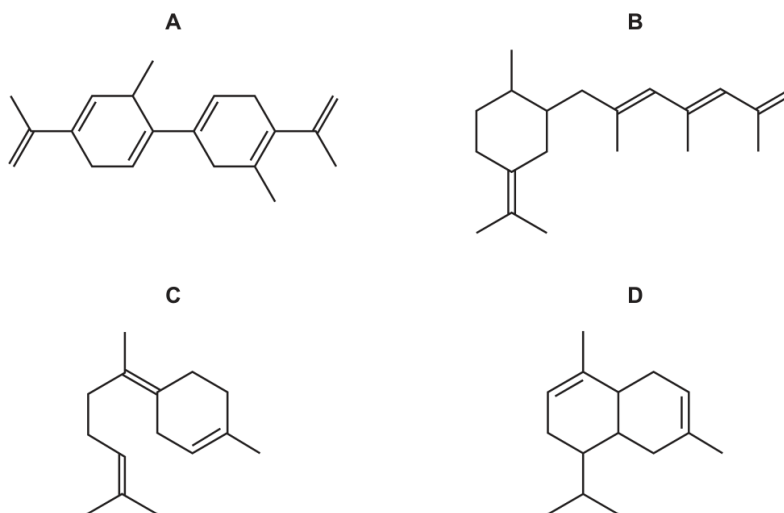
0.833 g of the carbonate releases 200 cm^3 of gas, measured under room conditions.

What is the identity of the metal carbonate?

- A Na_2CO_3 B K_2CO_3
C MgCO_3 D CaCO_3

[1]

12. When 0.010 mol of a hydrocarbon X reacts with 720 cm^3 of hydrogen at room conditions, an alkane is formed. What is hydrocarbon X?



[1]

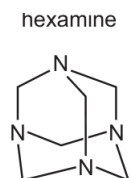
Earlier skills · mole, empirical, hydrate, limiting

13. How many molecules are present in 62 g of solid white phosphorus, P_4 ?

- A L B $2L$
C $\frac{L}{2}$ D $\frac{L}{4}$

[1]

14. Hexamine is a crystalline solid used as a fuel in portable stoves. The diagram shows its skeletal structure.



What is the empirical formula of hexamine?

A CH_2N

B $\text{C}_3\text{H}_6\text{N}_2$

C $\text{C}_4\text{H}_8\text{N}_4$

D $\text{C}_6\text{H}_{12}\text{N}_4$

[1]

15. Barium dithionate, $\text{BaS}_2\text{O}_6 \cdot 2\text{H}_2\text{O}$, is soluble in water. $\text{S}_2\text{O}_6^{2-}$ ions slowly decompose in acidic solution.



3.513 g of $\text{BaS}_2\text{O}_6 \cdot 2\text{H}_2\text{O}$ is dissolved in water in a 100 cm^3 volumetric flask and the solution made up to the mark with $\text{HCl}(\text{aq})$.

At time x min, a white precipitate of mass 0.661 g is present in the flask.

What is the concentration of BaS_2O_6 in the volumetric flask at time x min?

A $0.0077\text{ mol dm}^{-3}$

B $0.0090\text{ mol dm}^{-3}$

C 0.077 mol dm^{-3}

D 0.090 mol dm^{-3}

[1]

16. An excess of chlorine was bubbled into 100 cm^3 of hot 6.0 mol dm^{-3} sodium hydroxide.

How many moles of sodium chloride would be produced in the reaction?

A 0.30

B 0.50

C 0.60

D 0.72

[1]